

Introduction

Power BI Desktop is the free tool from Microsoft for developing Business Intelligence solutions. This tutorial walks you through importing and manipulating data with the Query Editor in Power BI Desktop.

This tutorial focuses specifically on Power BI Desktop because of the important role it plays in creating reports that can then be used by the other components.

Before starting this tutorial, make sure you have Power BI installed on your machine. If you are using a work laptop, the software might have already been installed. If you have your laptop, follow the instruction [here](#) to download and install the software.

The features in Power BI Desktop are organised into three views that you access from the navigation pane on the left side of the main window:



- **Report view:** A canvas for building and viewing reports based on the datasets defined in *Data* view.
- **Data view:** Defined datasets based on data retrieved from one or more data sources. *Data* view offers limited transformation features, with many more capabilities available through Query Editor, which opens in a separate window.
- **Relationships view:** Identified relationships between the datasets defined in *Data* view. When possible, Power BI Desktop identifies the relationships automatically, but you can also define them manually.

To access any of the three views, click the applicable button in the left navigation pane. When working in Power BI Desktop, you will normally perform the following four basic steps, although not necessarily in a single session:

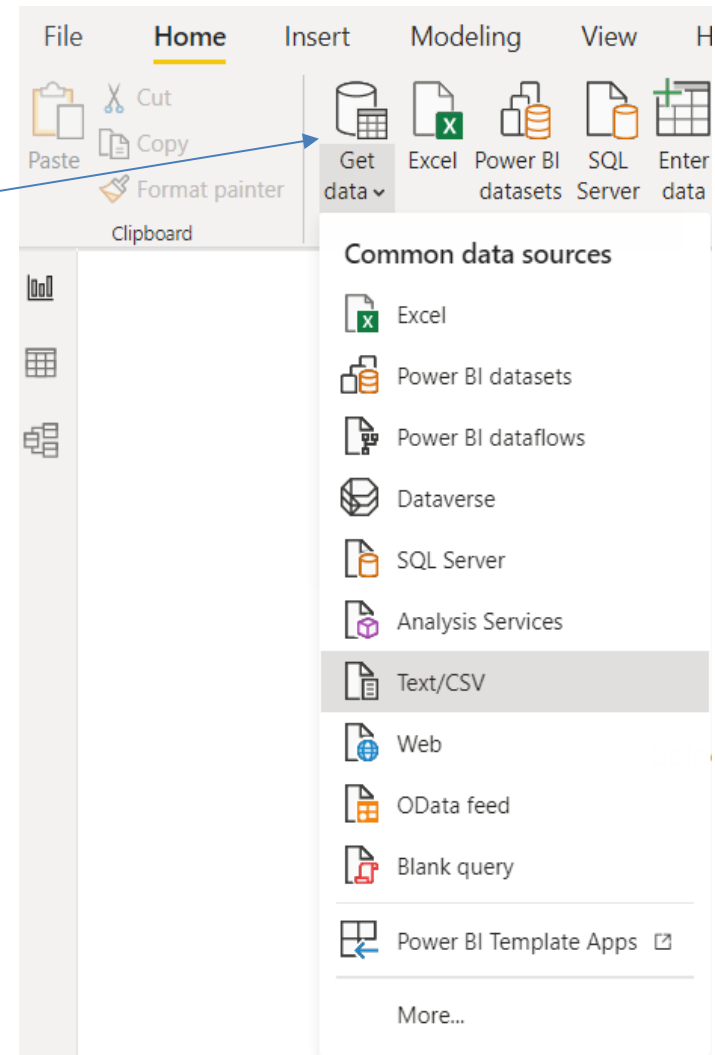
1. Connect to one or more data sources and retrieve the necessary data.
2. Transform and enhance the data using *Data* view, *Relationships* view, or the Query Editor, as necessary.
3. Create reports based on the transformed data, using *Report* view.
4. Publish the reports to the Power BI service or upload them to Power BI Report Server.

This tutorial will give you a better sense of how to retrieve and prepare the data, which is essential to building comprehensive reports. You will also learn more about building reports with different types of visualizations, but first, you need to make sure you get the data right.

Step 1: Getting the data in Power BI

This first session offers several examples of how to choose the right chart for each use case and how to improve them using some best data visualization principles. The visualizations are based on a single dataset from the data in *superstore_dataset_2011-2015*. Please download this file from Blackboard.

The Get Data tool can be used to import the dataset. To do so, click the drop-down under Get Data and select Text/CSV to load the data. Browse the local computer and select the file to be loaded.



superstore_dataset_2011-2015.csv

File Origin

1252: Western European (Windows)

Delimiter

Comma

Data Type Detection

Based on first 200 rows

Extract Table Using Examples

Load

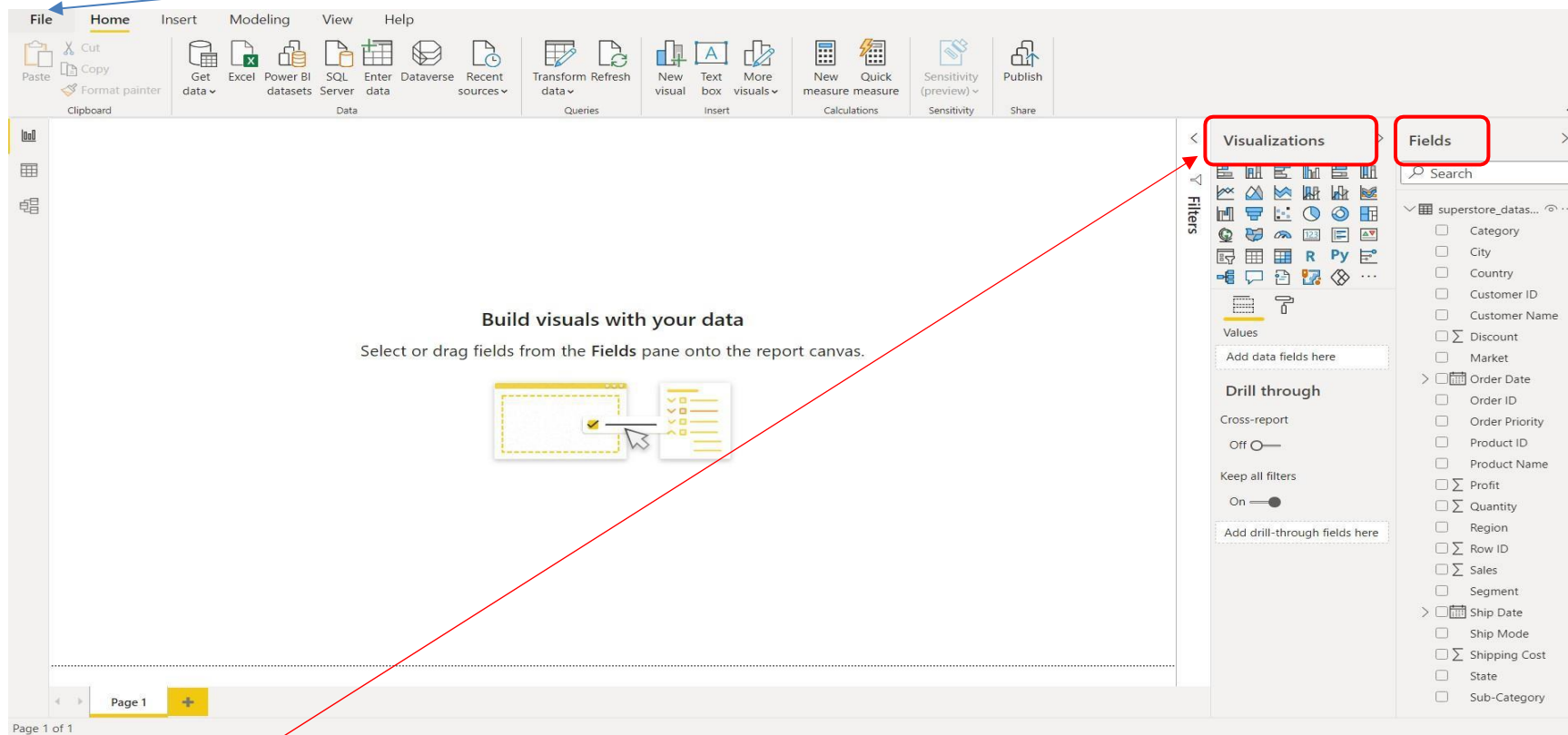
Transform Data

Cancel

Once the file is selected, click *Load* to import the data into the Power BI desktop. After the data has loaded, go to *Data* view to verify whether the dataset has been imported.

Step 2: Getting started with some Visualizations

With the dataset in place, visualizations can now be added to the report. But before you do, save the file as “My Report” if you have not already done so. This can be done by clicking on **File** -> **Save**.



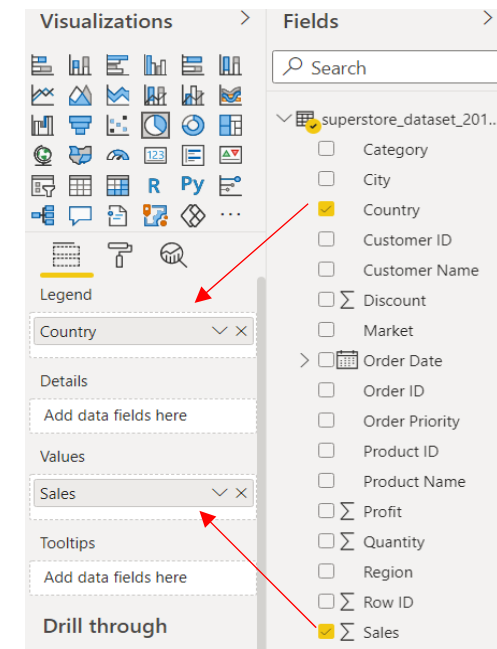
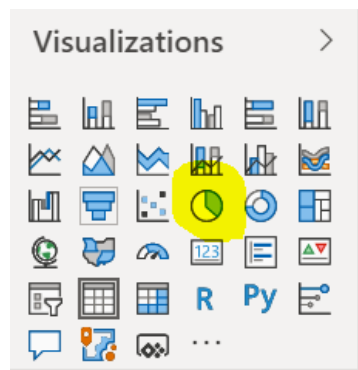
The **Visualizations** pane provides the tools necessary for adding and configuring the visualizations. The **Fields** pane includes a list of the defined datasets, with access to each dataset column. The above figure indicates that the *superstore_dataset* is the only visible dataset included in this report file. In the next steps, let us start with some activities.

Activity 1: Creating a chart to visualize the top countries with the highest sales

In this section, you will start exploring different visuals. The visualizations will first display sales information for all countries, then it must be filtered to show only a few countries. Although many different types of visualizations will be created, all are configured the same way.

Let us start with *Pie chart* visual, take the following steps:

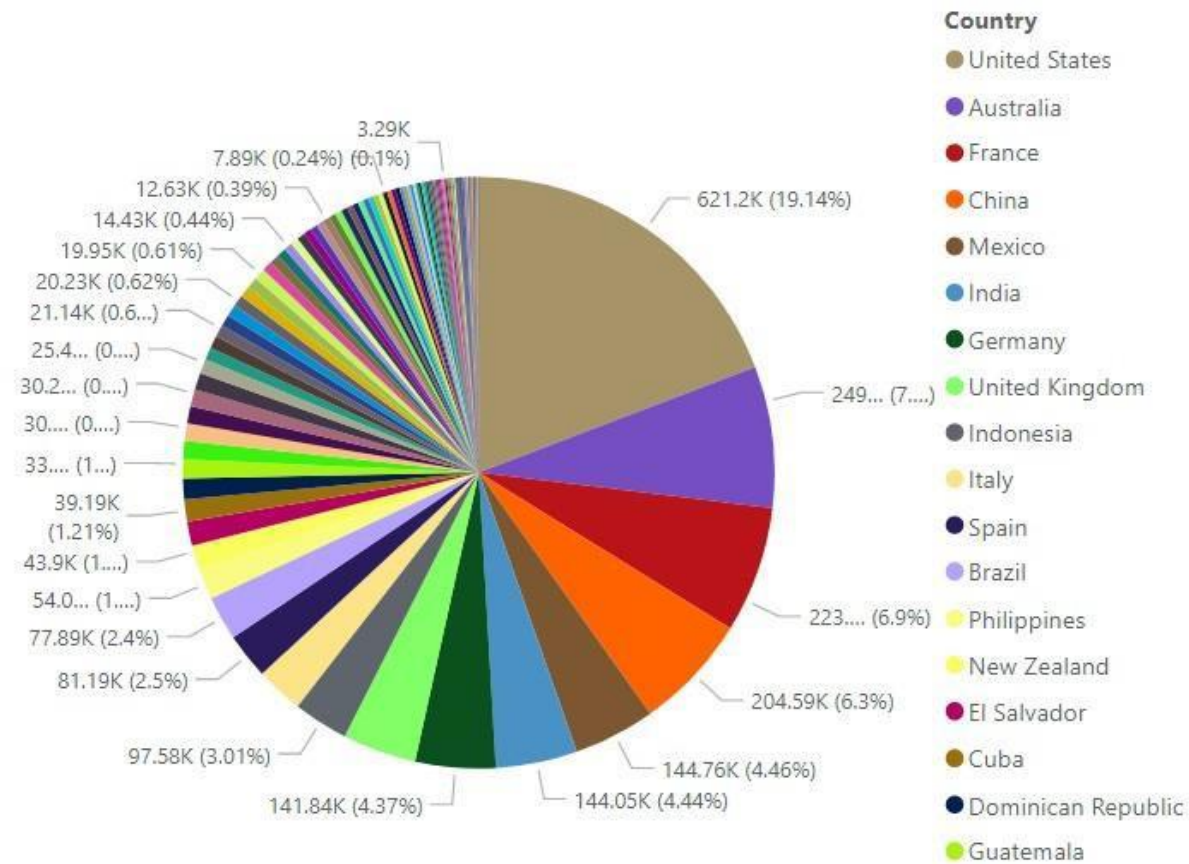
1. Click the *Pie chart* button on the Visualization pane, as shown below.



2. Drag the *Country* column from the *Fields* pane to the *Legend* section of the *Visualizations* pane (see example on the right).
3. Drag the *Sales* column from the *Fields* pane to the *Values* section of the *Visualizations* pane.

The resulting pie-chart is shown below:

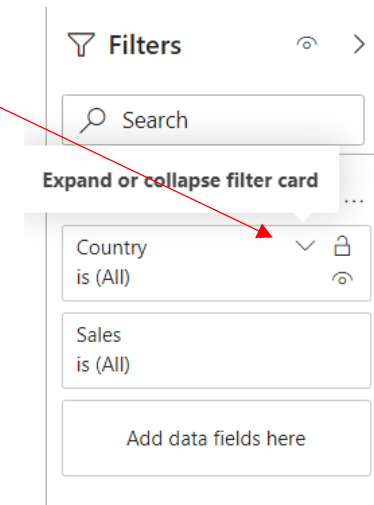
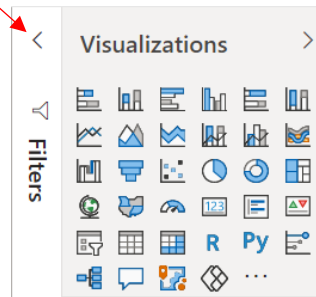
Sales by Country



This chart looks clumsy and confusing at this point but we are not finished yet. In the next step, let us filter for just the top six countries.

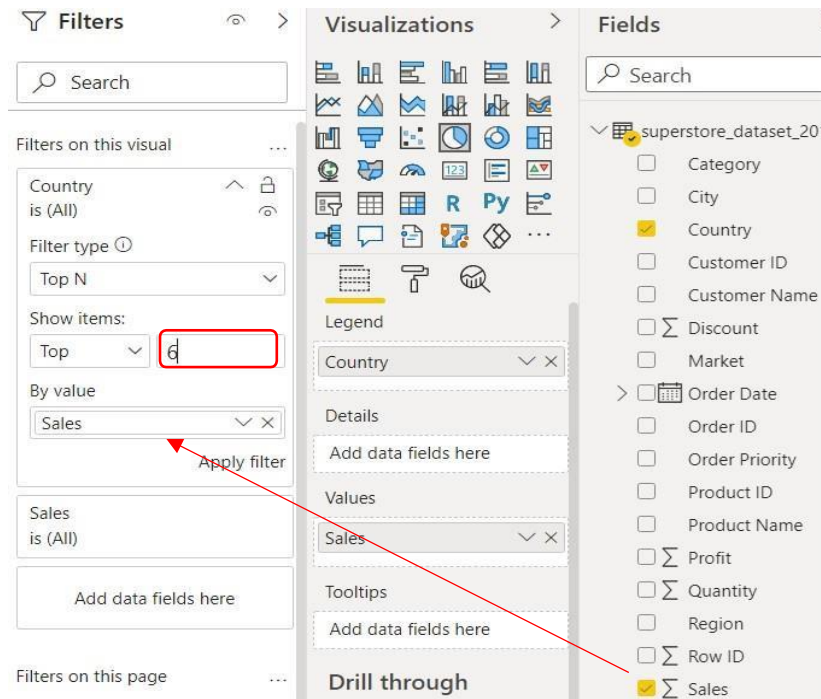
4. Expand the *Country* section in the *Filters* section by clicking on the drop-down menu here.

NOTE: If you do not see the *Filters* section, **click** on the small arrow mark here.



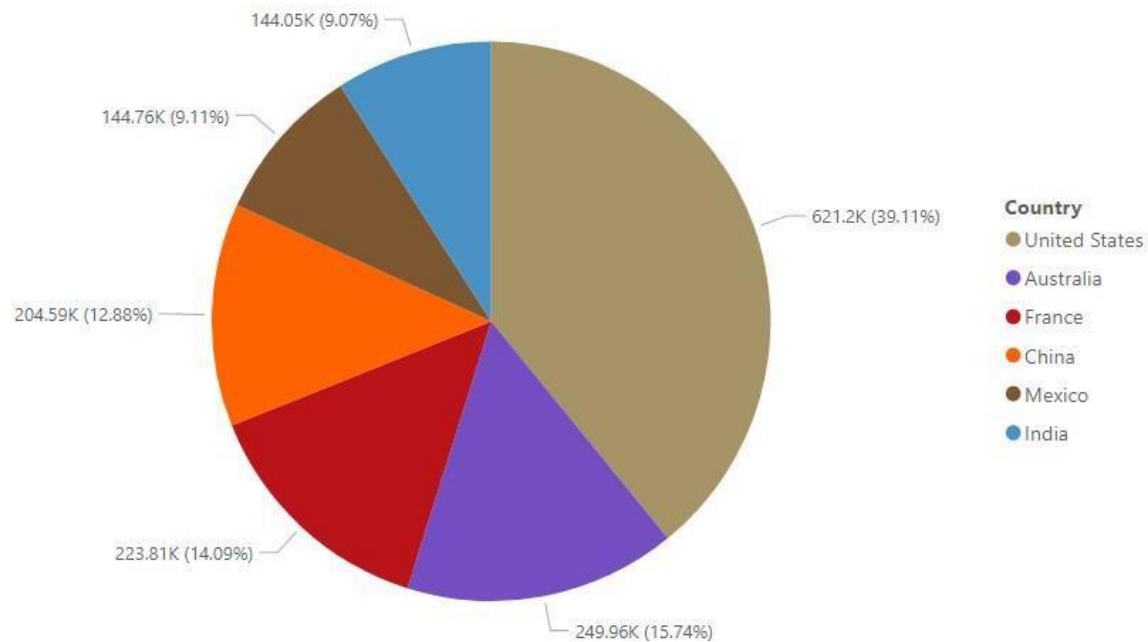
5. Change the *Filter Type* to 'Top N'. Then, enter the value '6' in the adjacent field. (see below chart)

6. Drag and drop the *Sales* column into *By Value* field. Click *Apply Filter*.



The final pie-chart is shown below:

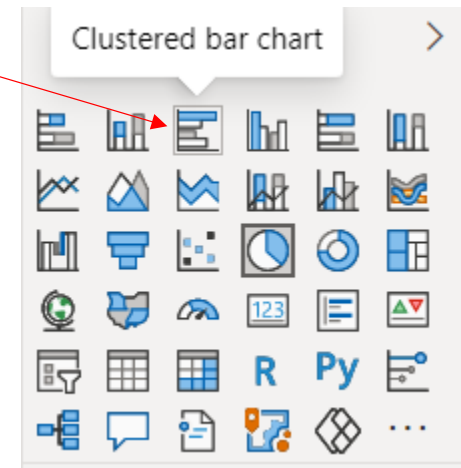
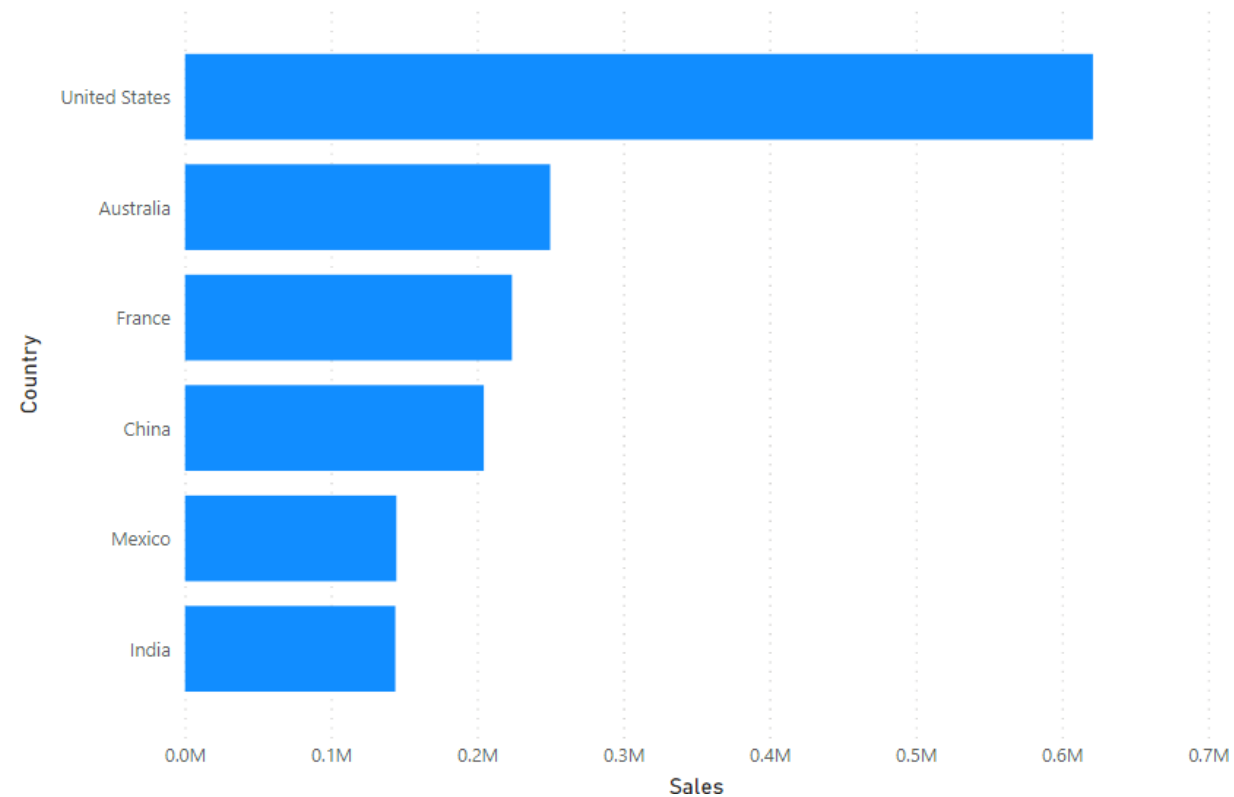
Sales by Country



Now, the chart looks good as it clearly shows the sales proportions of the top six countries. However, if we want to quickly find out the country with the third highest sales, it cannot be identified easily. So, let us try to visualize using another chart.

7. Ensure that the *Pie chart* visual **is selected** (click on it to make sure it is selected). Then, **select** the Clustered bar chart. Visualization will now look like below.

Sales by Country

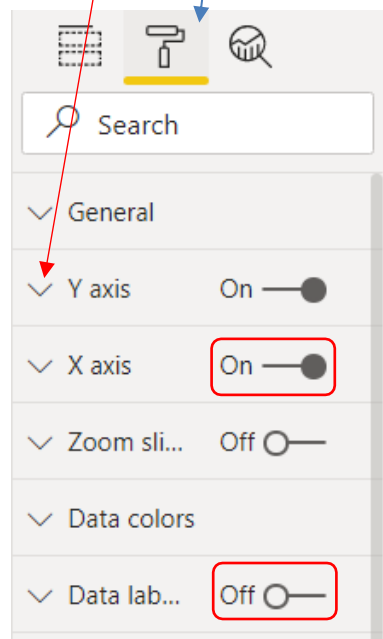


In the next steps, let us do some formatting like changing the title, adding data labels, etc. to make the chart look better.

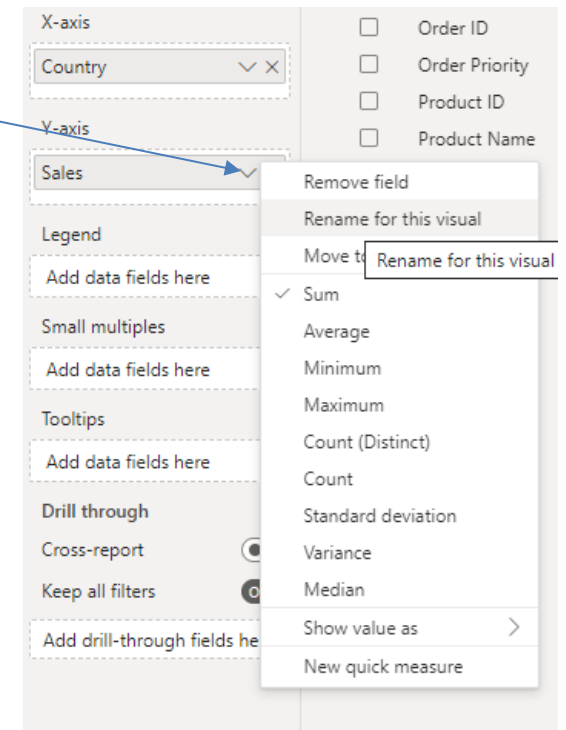
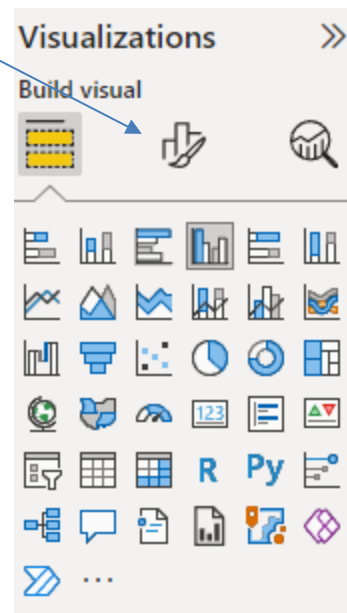
8. In the *Visualisation* pane, the drop-down adjacent to *Sales* column in the *Values* field must be clicked. Then, the option 'Rename for this visual' must be selected.

9. In the next step, rename the *Sales* into *Amount in Dollars*

10. Select the *Format* tab, change the X axis option to 'Off' and Data Labels to 'On'.
Then, **click** on the drop-down adjacent to Y axis.



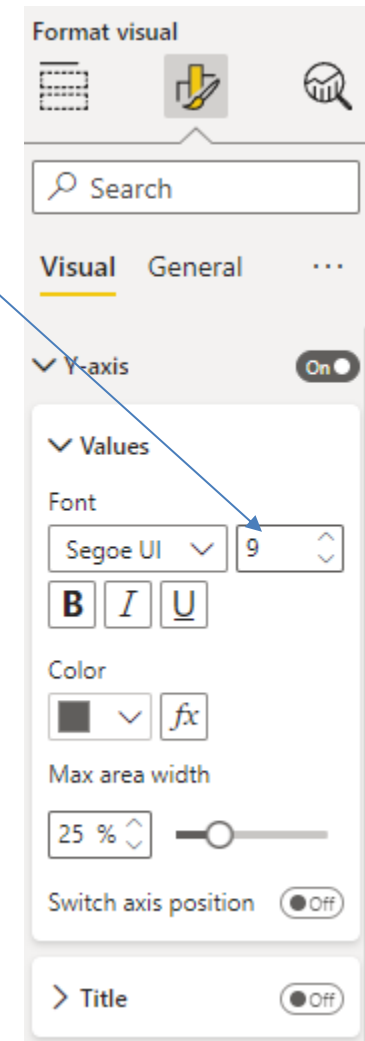
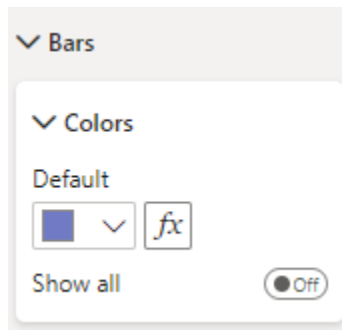
If you are using the **latest version of Power BI**, the format tab can be found at the top of the Visualisation pane



11. Under the *Y axis* field, change the *Text size* from '9' to '11' (see to the right). After that, slowly scroll under the same *Y axis* and change the *Title* to off (see below). *Y axis* field can be minimized now.

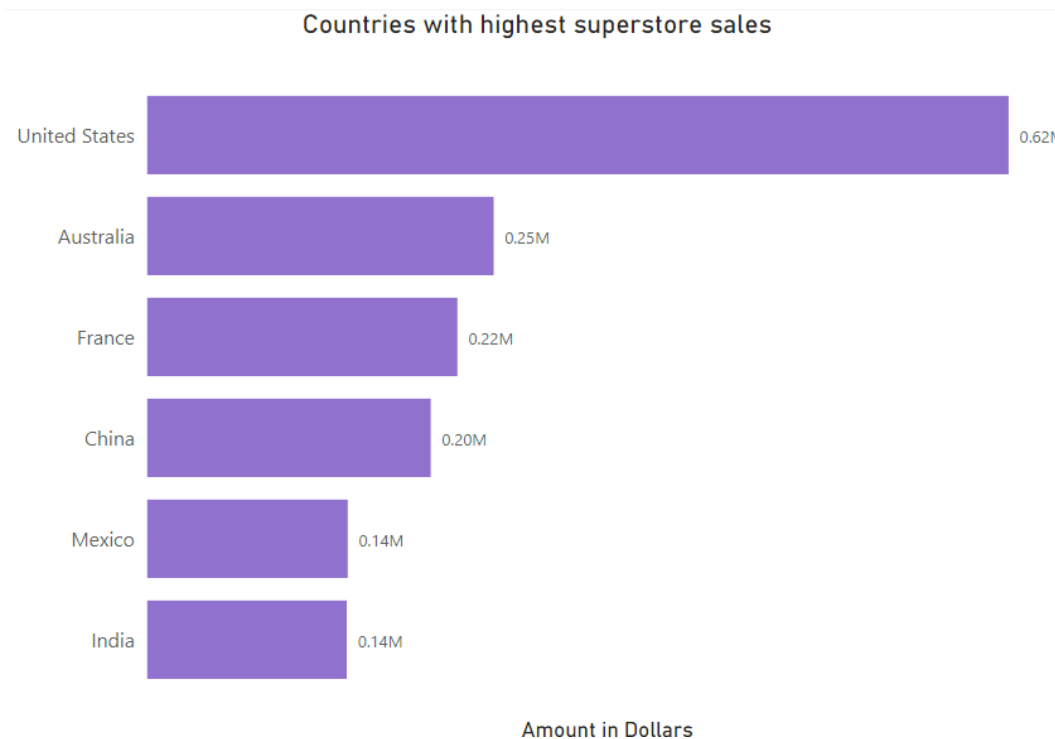
12. Moving on, scroll down until you see the *Bars* field (see below). Click on the drop-down next to *Default color* and choose other color as per your choice. Here, we have selected a purple shade. After choosing the color, minimize the *Bars* field.

NOTE: If you want to pick your own color, please visit the website [Color Picker](#) and find the hex code for your color (use the More Colours Option to add the hex code).



13. Let us move on to the final change for this visual. Click on the "General" tab and Scroll down to the *Title* field. Change the *Title text* to 'Countries with highest superstore sales'. Change the *Alignment* to centre. Finally, increase the *Text Size* to 16.

Finally, the finished chart looks like this. (see below)



Visualizations

Format visual

Search

Visual **General** ...

> Properties

▼ Title On

Text

Countries with the h fx

Heading

Heading 3

Font

DIN 16

B *I* U

Text

■ ▼ fx

Background

□ ▼ fx

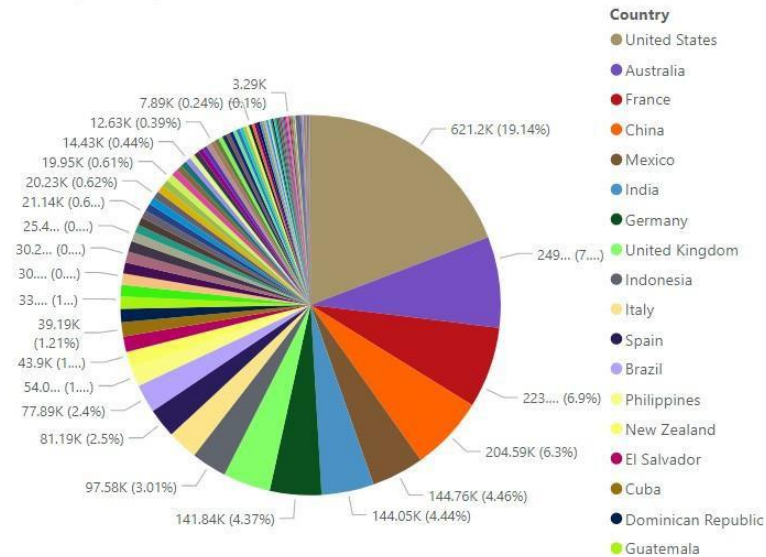
Horizontal alignment

☰ ☷ ☶

Text wrap On

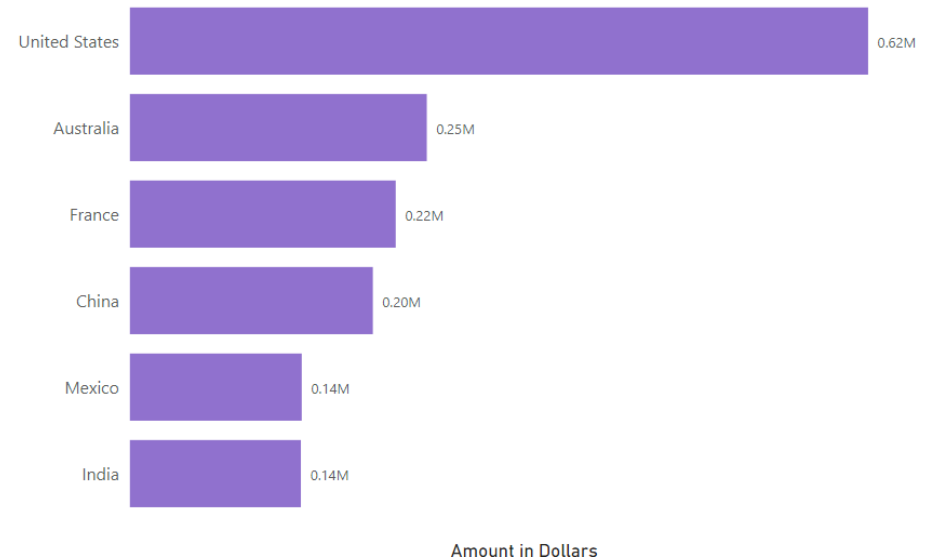
Let us rewind to our initial chart and compare it with the final chart we came up with.

Sales by Country



Before

Countries with highest superstore sales



After

Here, we can clearly see the advantages of some best practices and the data visualization principles in full effect. It is advisable to know how to make our charts look better and easily understandable.

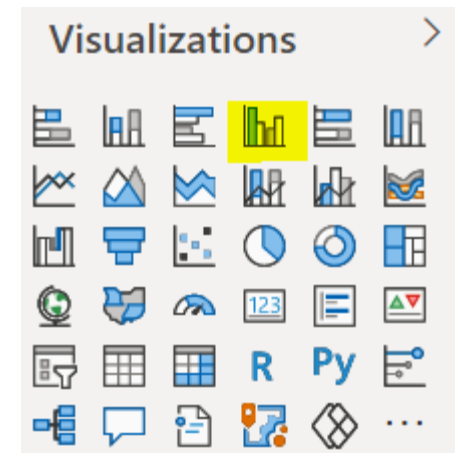
Activity 2: Creating a chart to compare the categorical profit over years

In this second tutorial, the requirement will be to perform a comparison study. So, let us start with a chart where it is easy to do multiple categorical comparisons.

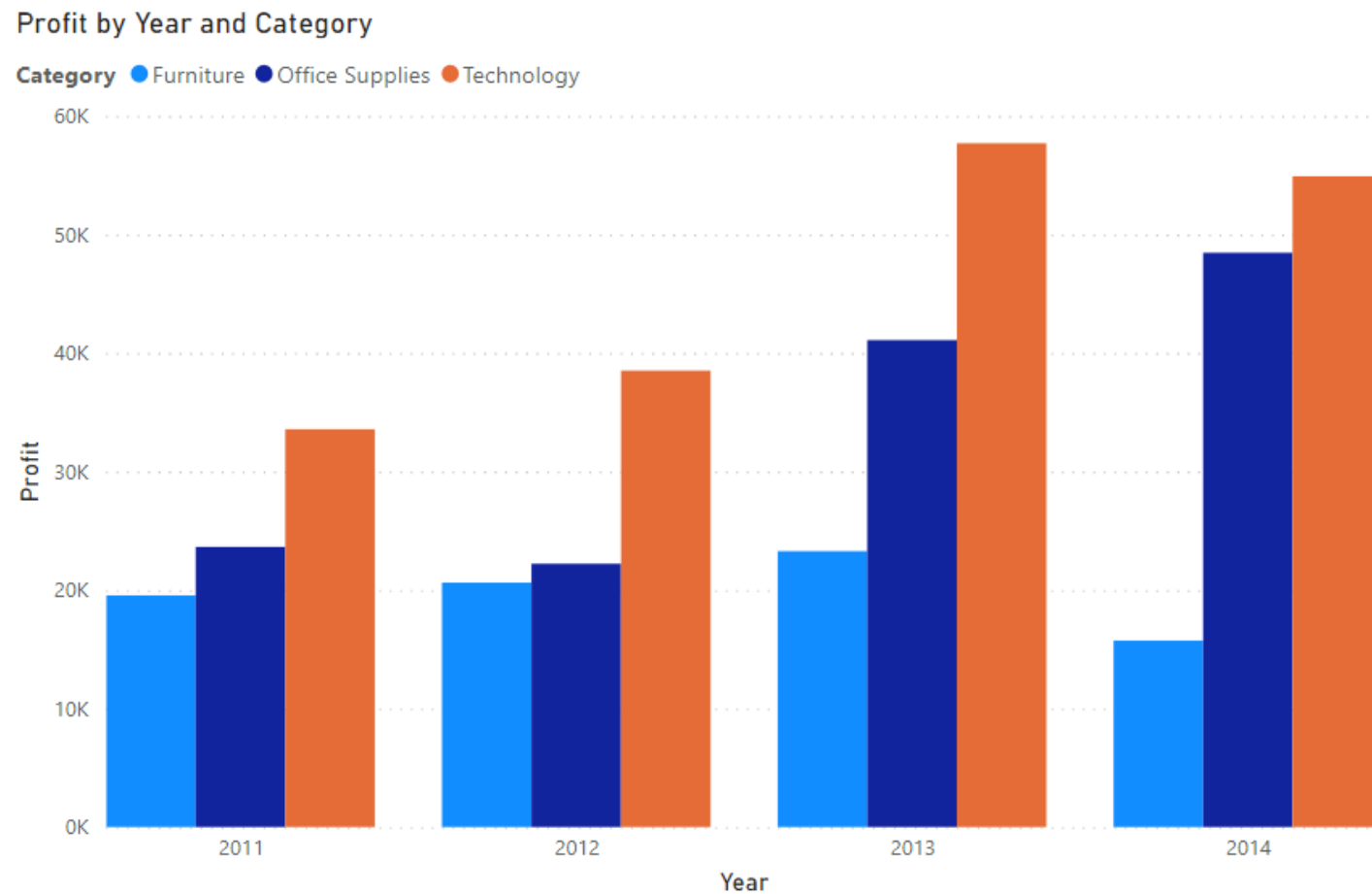
1. Before starting, you can create a new page for this visual by clicking on the plus mark at the bottom of the page adjacent to 'Page 1'.

Or you can keep working on the same page by clicking in the white area outside your chart to deselect it.

2. Click the 'Clustered Column Chart' on the Visualizations pane. (See to the right)
3. Drag the *Order Date* column from the *Fields* pane to the X-axis section of the *Visualizations* pane. The axis provides the foundation on which the chart is built.
4. Drag the *Category* column from the *Fields* pane to the *Legend* section of the *Visualizations* pane. The legend determines how data will be clustered.
5. Drag the *Profit* column from the *Fields* pane to the Y-axis section of the *Visualizations* pane. The value serves as the measure used for aggregating the data.

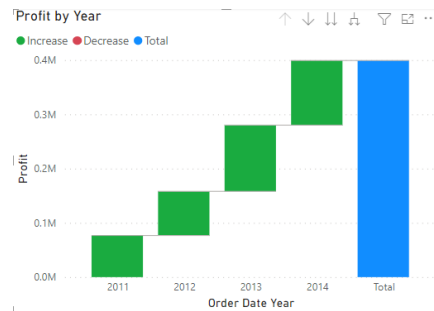


The chart so far looks like this.

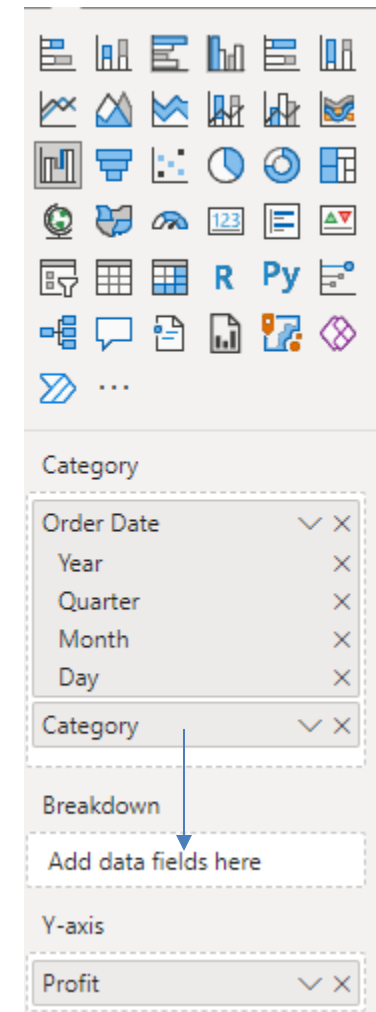
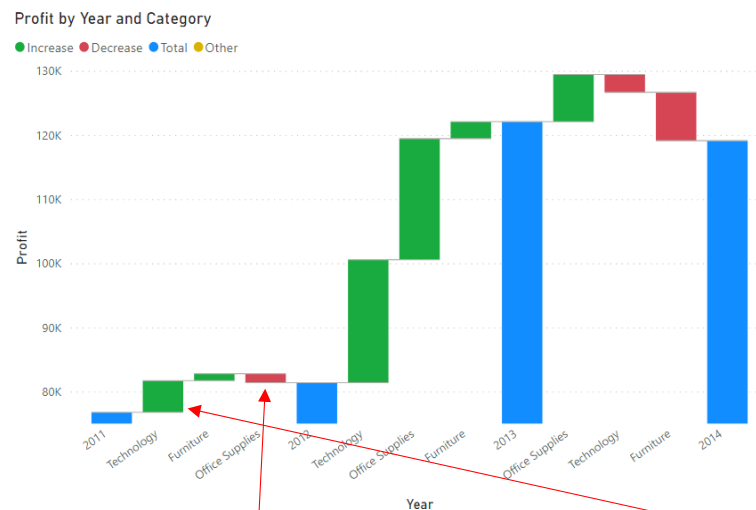


This chart is not bad, but it is hard to read the comparison on a yearly basis. Let's try some other chart where the difference in magnitude for each category on yearly basis is easily displayed.

- Ensure that the *Clustered Column Chart* chart visual **is selected** (click on it to make sure it is selected). Then, **select** the Waterfall Chart from the *Visualizations* pane. The first icon one in the third row. (see to the right) A waterfall or cascade chart is used to show changes in revenue or profit between two time periods.



- Select and drag the *Category* column into the *Breakdown* field. This has to be done so that the breakdown can be seen by different categories. Visualization will now look like below.

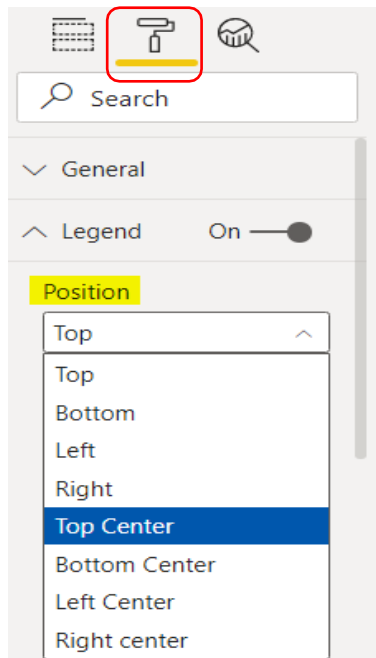
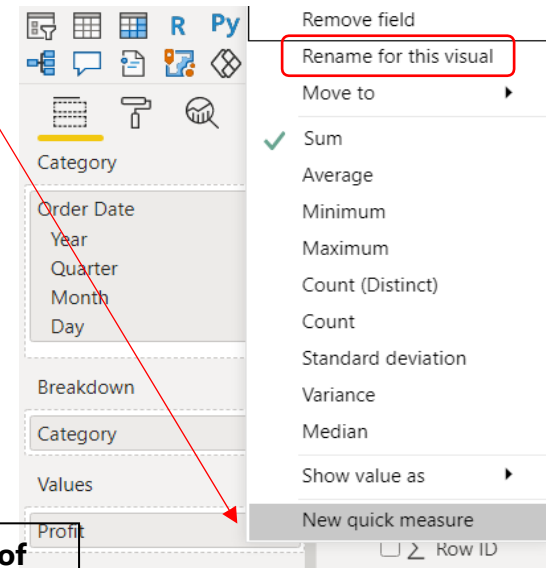


In the chart, it can be seen that when compared to year 2011, the superstore in the year 2012 made **profit** in the Technology as well as Furniture category but suffered some **loss** in the Office Supplies category.

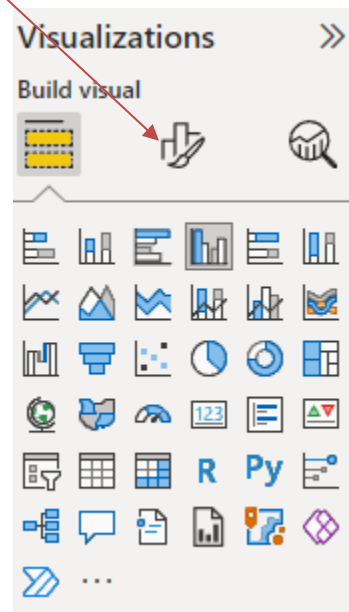
8. In the *Visualisation* pane, the drop-down adjacent to *Profit* column in the *Values* field must be clicked. Then, the option 'Rename for this visual' must be selected.

9. Rename the text *Profit* to '*Amount in Dollars*'. The formatting for the chart will be done in the upcoming steps.

10. Start by selecting the *Format* tab, then select the *Legend* field. Change the *Position* To '*Top Centre*'. Next, minimize *Legend* field.

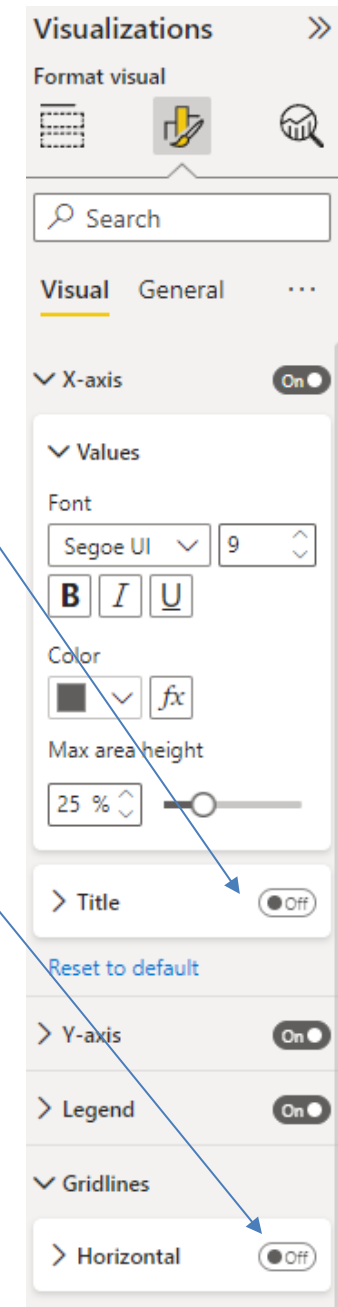


Remember that if you are using the **latest version of Power BI**, the format tab can be found at the top of the Visualisation pane



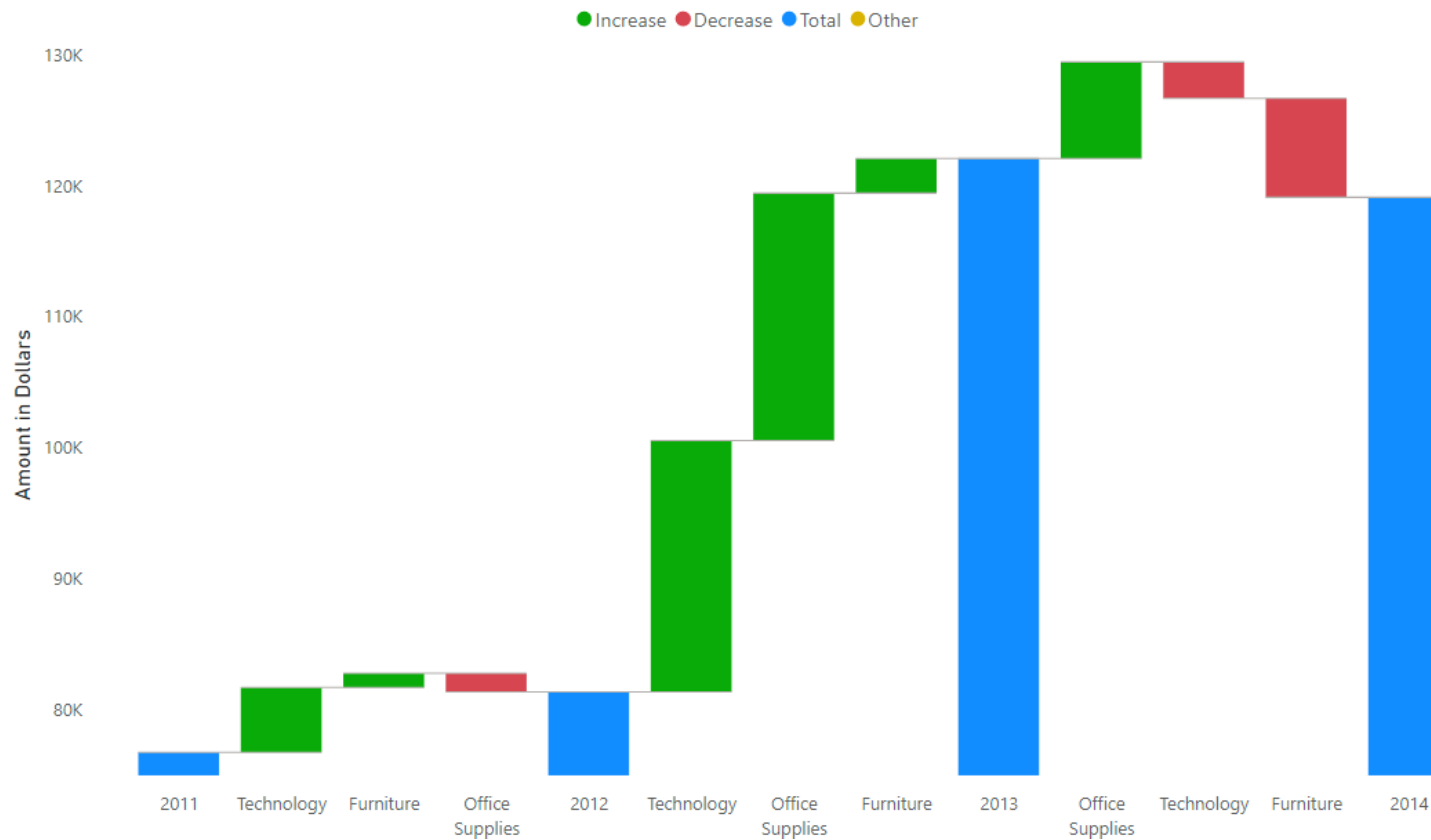
11. Select the *X axis* field (see to the right). Then, slowly scroll down and change the *Title* to 'Off'. (see below)
Minimize the *X axis* field by clicking the **arrow** mark.

12. Select the *Y axis* field (see to the right). Then, slowly scroll down and change the Gridlines to 'Off'. (see below)



Up to step 12, the chart looks something like this. If your chart does not look like the below chart, go back and check whether you have missed any steps.

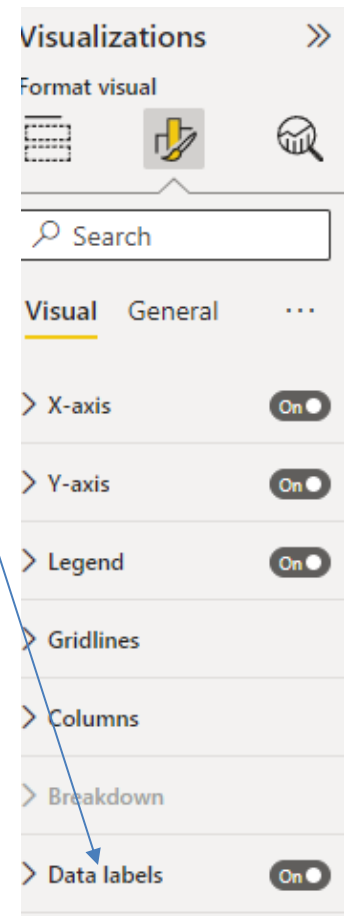
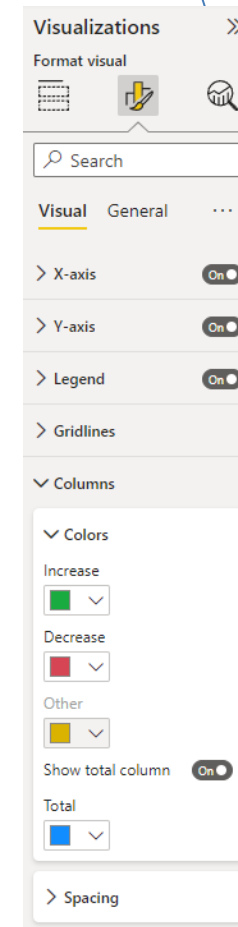
Profit by Year and Category



The formatting is still not finished. We will see how to change the colors of the bars, title, add some textual elements in the coming steps.

13. Select the Data label field (see to the right). Turn it 'On'. This feature is necessary to see the data labels on top of the bars. The values displayed on top of the **year** column are the summation of amount for the three categories (Technology, Furniture, Office Supplies). The values displayed on top of the **category** column is the amount calculated by taking the difference between the next year and the current year,

14. Scroll down until you find the *Columns* colors field. Click on it and then select the green color under the *Increase* field. Here, we are trying to change some colors so that their meanings are easily understood.



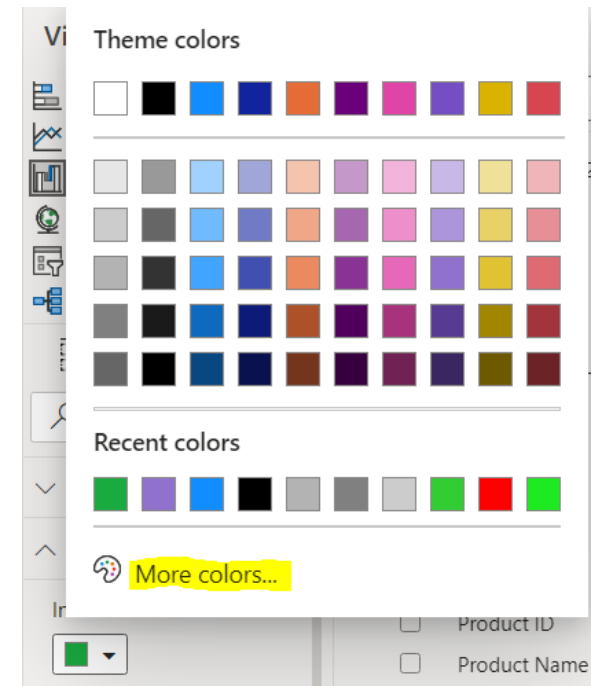
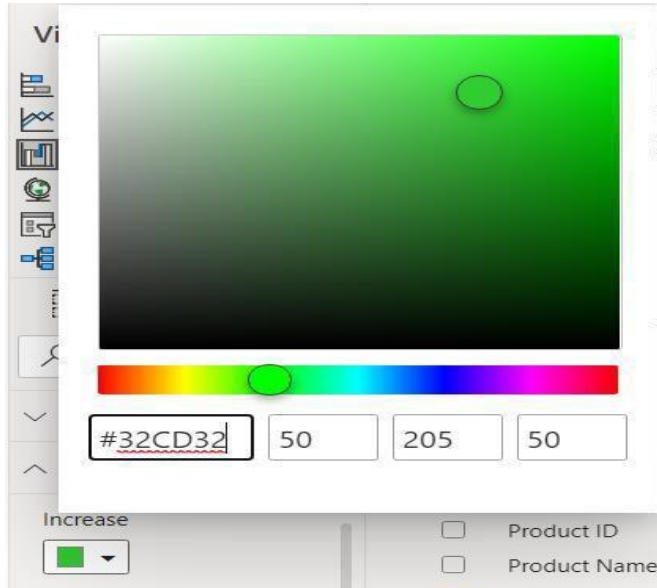
15. After clicking the green color as mentioned in previous step, a pane with different colors is shown. Next, click on More colors... (see to the right)

16. Change the color to lime green (hex code #32CD32). Similarly, change the color for the other fields under the Sentiment colors.

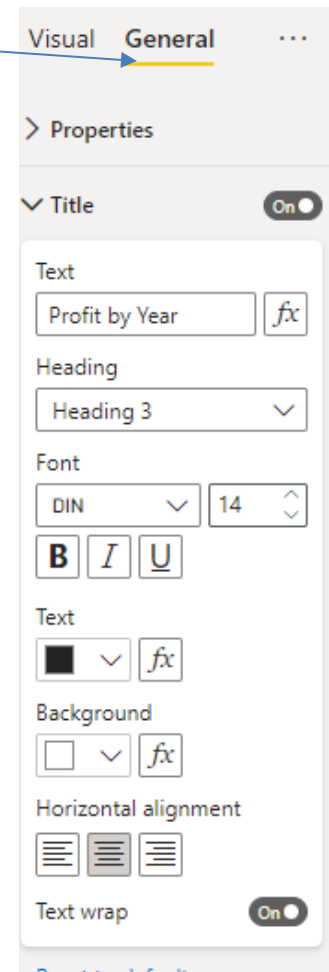
Decrease ----- > #FF0000

Other----- > #000000

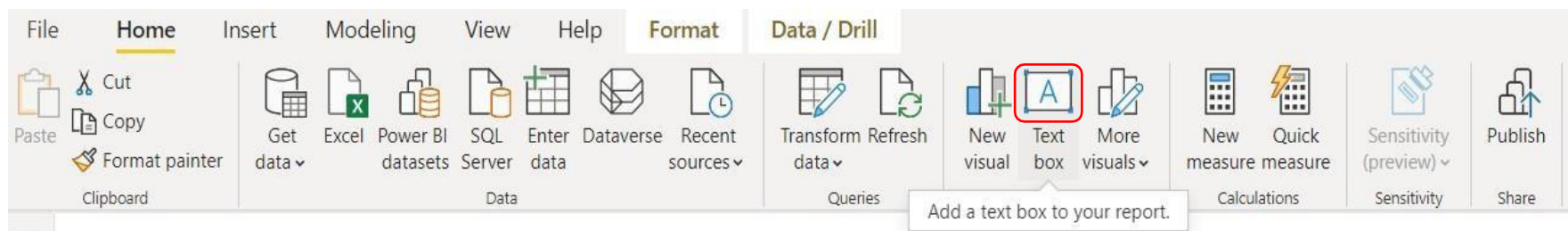
Total ----- > #B3B3B3



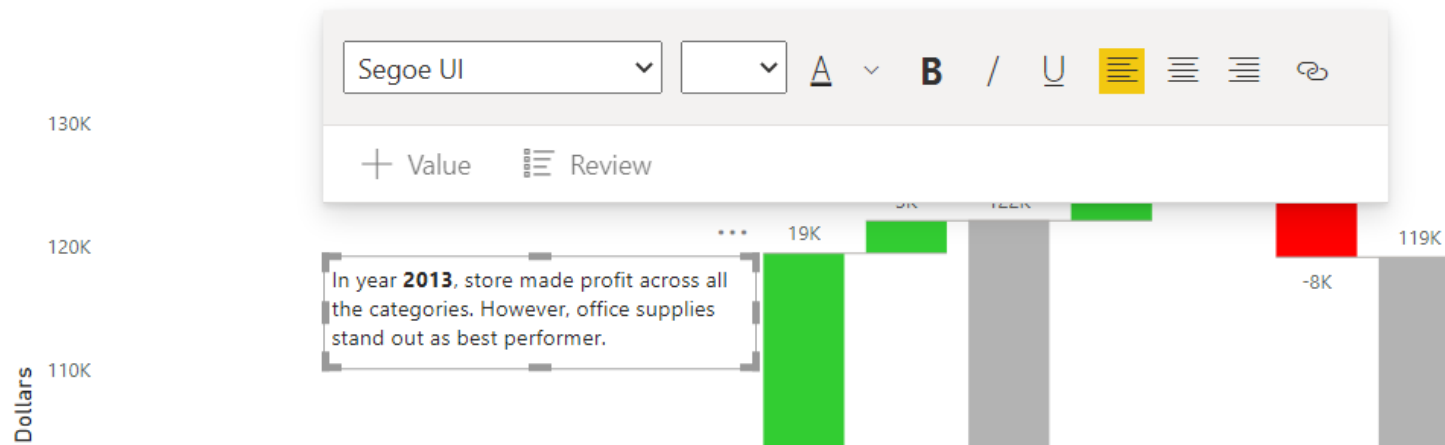
14. After all the colors are set, there is just one step left in the formatting section. Switch to the General tab and find the *Title* of the chart. Change the *Title text* to 'Yearly Profit/Loss Review Across Categories'. Change the *Alignment* to centre. Finally, increase the *Text Size* to 16.



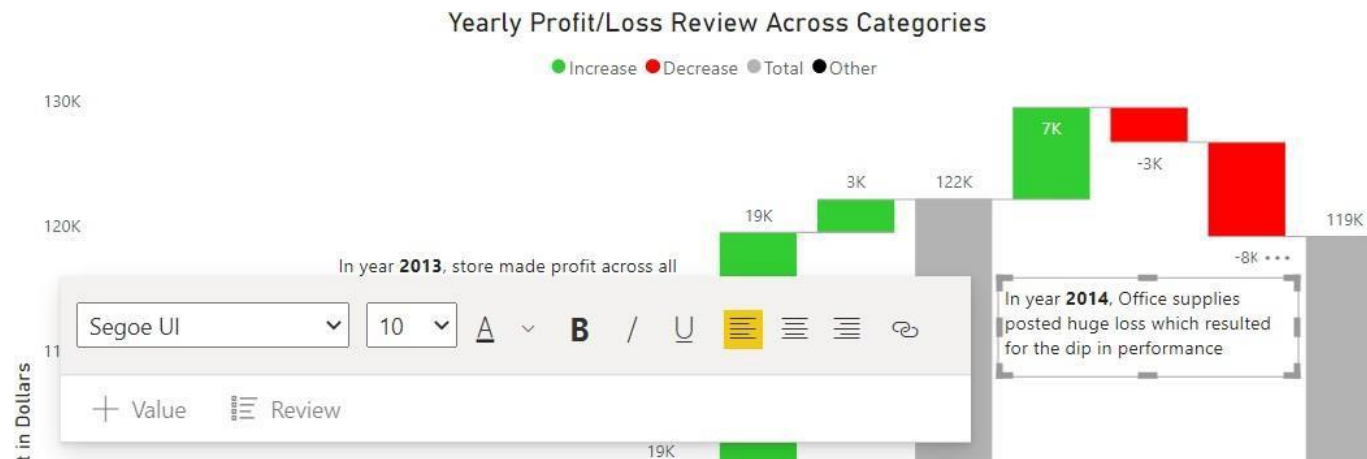
15. The text box is used to add text elements to the chart. This is useful to tell the story for a chart and make the chart easily understandable for the audience. At the top, select *Text Box* to add to your report. (see below)



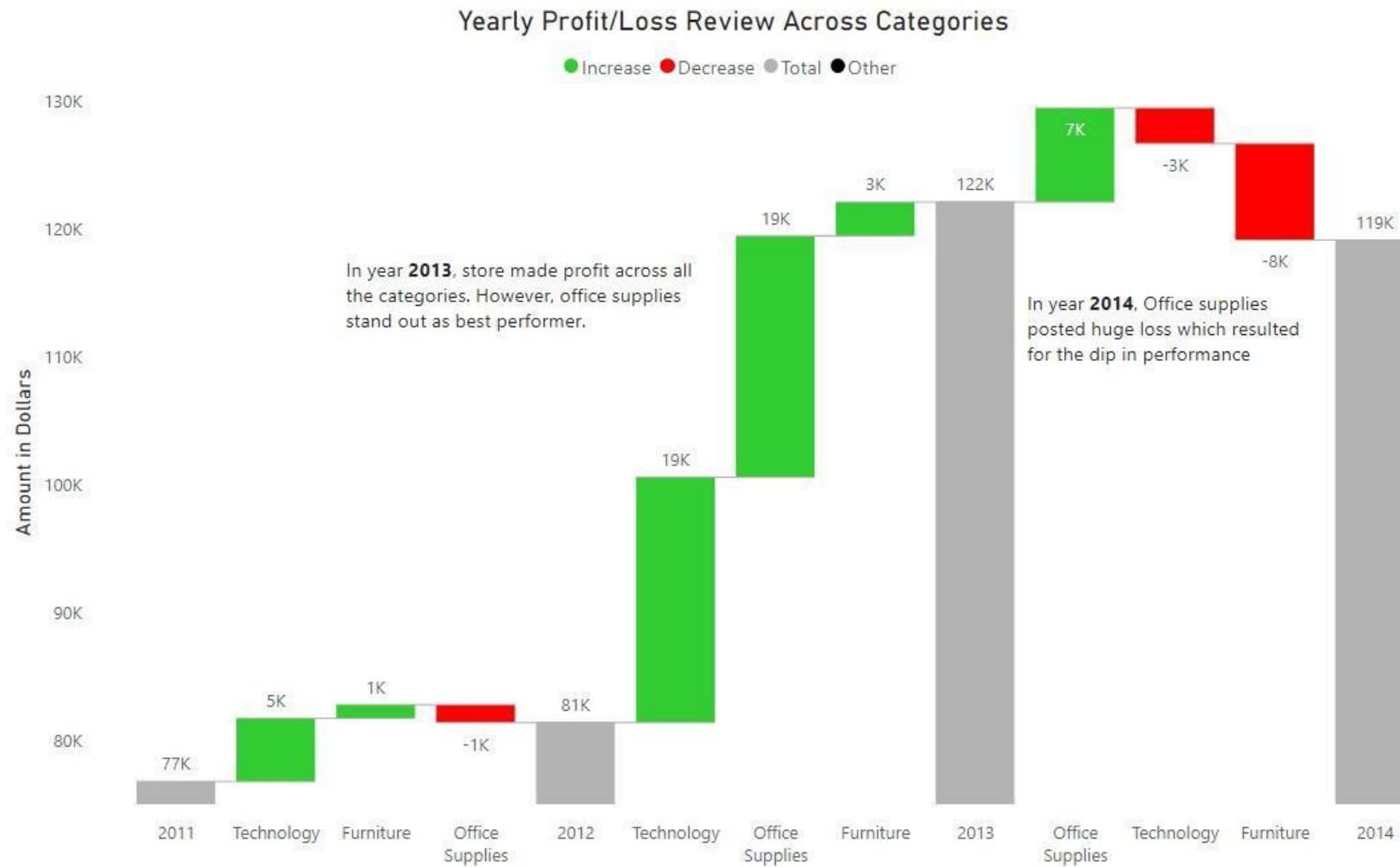
16. Reshape the *Text Box* as per the requirement and carefully place it one the chart as shown below.



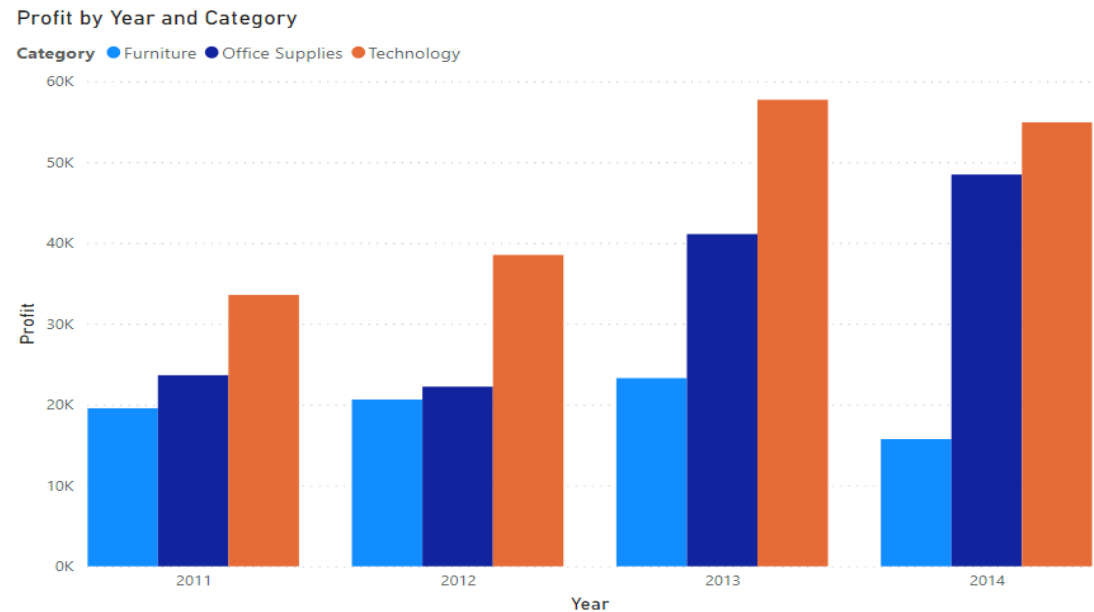
17. Similarly, add one more *Text Box*. Add the text and place it on the chart as shown below.



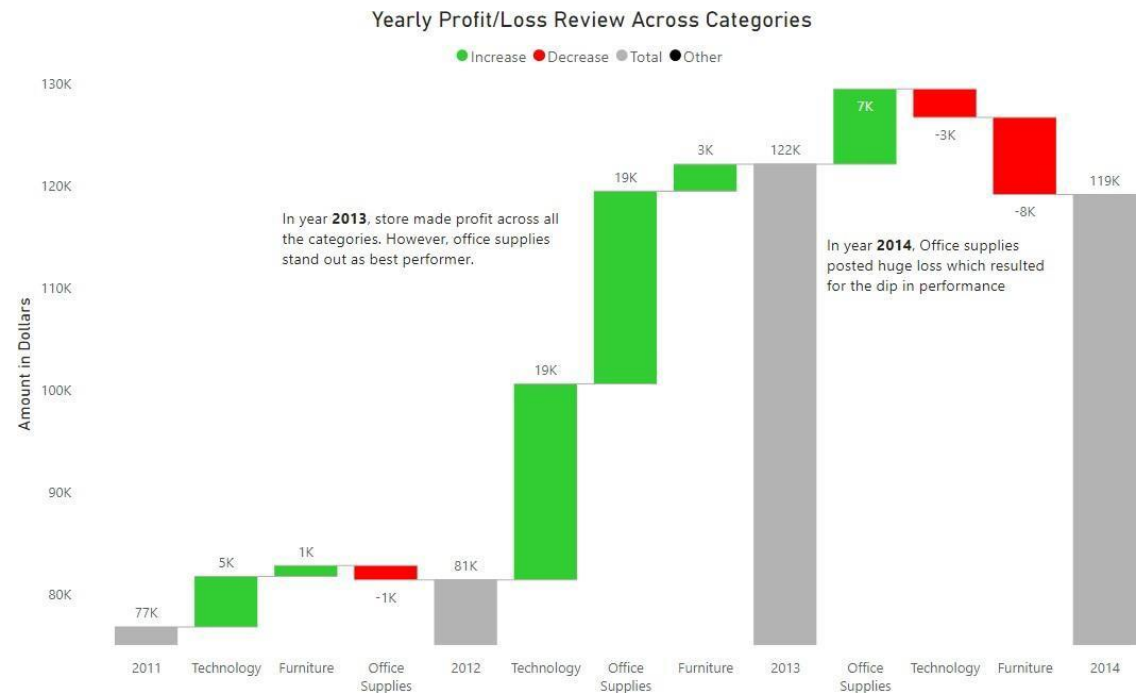
Finally, the chart looks like this after all the changes.



BEFORE: Clustered Column chart to visualize the differences of profit over various categories. This chart is good to see the generic trend but not good enough to accurately measure the changes. So, if we know the right chart to use as per the use case, then the story can be told better.



AFTER: Waterfall chart to visualize the changes in profit between two time periods. In this chart, with the addition of textual elements, the reasons for the good and bad performances can be identified easily along with the magnitude of changes.



Activity 3: It's your turn now!

Now it's your turn! In this last task, you will be able to implement what you have learnt so far in the previous activities.

Please download the Covid dataset from blackboard and create a chart to visualize the "weekly deaths due to covid-19 compared to other causes".

The resulting chart should help us understand the general trend over time and do a comparison between the two groups (number of deaths for covid versus other causes).

If you are stuck somewhere, please ask your tutor.

You will also find the solution on Blackboard next week!

(ActivitySolution File).

Don't forget to share your work on Padlet (link on Blackboard)!

At the end of this module, the best post will be selected for the "Tony Ruff – Power BI Award"! Good luck!